Deliverable 3 - INF201

Authors: Amena and Arthika

GitHub: https://github.com/amhus8955/INF201_deliverable_3

In [1]:
```python
import random

from battle import Battle
from character import Character, Mage, Warrior
from dice import Dice
from items import Weapon

random.seed(42)  # reproducible results


def present(obj) -> None:
    """Show the attributes and public methods of an object's class."""
    cls = type(obj)
    print(f"{cls.__name__} attributes: {vars(obj)}")
    print(f"{cls.__name__} methods: {[n for n in dir(cls) if not n.startswith('_')]}")
```

The classes

In [2]:
```python
sword = Weapon("Sword", Dice(sides=8))
present(Dice(sides=20))
present(sword)
present(Character("Hero"))
present(Warrior("Thorin"))
present(Mage("Gandalf"))
present(Battle(Warrior("Thorin"), Mage("Gandalf")))
```

```
Dice attributes: {'sides': 20}
Dice methods: ['roll']
Weapon attributes: {'name': 'Sword', 'damage_dice': Dice(sides=8)}
Weapon methods: ['roll_damage']
Character attributes: {'name': 'Hero', '_hp': 100, 'hp_max': 100, 'damage': 10, 'weapon': None}
Character methods: ['attack', 'heal', 'hp', 'introduce', 'is_alive', 'take_damage']
Warrior attributes: {'name': 'Thorin', '_hp': 100, 'hp_max': 100, 'damage': 10, 'weapon': None,
'armor': 5}
Warrior methods: ['attack', 'heal', 'hp', 'introduce', 'is_alive', 'take_damage']
Mage attributes: {'name': 'Gandalf', '_hp': 100, 'hp_max': 100, 'damage': 10, 'weapon': None, 'm
ana': 30, 'mana_max': 30}
Mage methods: ['MANA_RESTORED', 'SPELL_COST', 'attack', 'heal', 'hp', 'introduce', 'is_alive',
'take_damage']
Battle attributes: {'hero': <Character name=Thorin hp=100 damage=10 weapon=None>, 'enemy': <Char
acter name=Gandalf hp=100 damage=10 weapon=None>}
Battle methods: ['play_round', 'simulate_battle']
```

Battles

In [3]:
```python
Battle(Warrior("Thorin"), Mage("Gandalf")).simulate_battle();
```

```
Round 1: Thorin 85 HP, Gandalf 90 HP
Round 2: Thorin 70 HP, Gandalf 80 HP
Round 3: Thorin 55 HP, Gandalf 70 HP
Round 4: Thorin 50 HP, Gandalf 60 HP
Round 5: Thorin 45 HP, Gandalf 50 HP
Round 6: Thorin 30 HP, Gandalf 40 HP
Round 7: Thorin 25 HP, Gandalf 30 HP
Round 8: Thorin 20 HP, Gandalf 20 HP
Round 9: Thorin 5 HP, Gandalf 10 HP
Round 10: Thorin 5 HP, Gandalf 0 HP
Thorin wins after 10 rounds!
```

In [4]:
```python
axe = Weapon("Axe", Dice(sides=8))
staff = Weapon("Staff", Dice(sides=6))
Battle(Warrior("Thorin", weapon=axe), Mage("Gandalf", weapon=staff)).simulate_battle();
```

```
Round 1: Thorin 85 HP, Gandalf 98 HP
Round 2: Thorin 70 HP, Gandalf 97 HP
Round 3: Thorin 55 HP, Gandalf 92 HP
Round 4: Thorin 55 HP, Gandalf 88 HP
Round 5: Thorin 54 HP, Gandalf 85 HP
Round 6: Thorin 39 HP, Gandalf 83 HP
Round 7: Thorin 39 HP, Gandalf 81 HP
Round 8: Thorin 39 HP, Gandalf 74 HP
Round 9: Thorin 24 HP, Gandalf 73 HP
Round 10: Thorin 24 HP, Gandalf 71 HP
Round 11: Thorin 24 HP, Gandalf 67 HP
Round 12: Thorin 9 HP, Gandalf 66 HP
Round 13: Thorin 8 HP, Gandalf 62 HP
Round 14: Thorin 8 HP, Gandalf 55 HP
Round 15: Thorin 0 HP, Gandalf 47 HP
Gandalf wins after 15 rounds!
```

In [5]:
```
Battle(Character("Hero", weapon=sword), Character("Goblin", hp_max=60, damage=8)).simulate_batt
```

```
Round 1: Hero 92 HP, Goblin 55 HP
Round 2: Hero 84 HP, Goblin 54 HP
Round 3: Hero 76 HP, Goblin 51 HP
Round 4: Hero 68 HP, Goblin 44 HP
Round 5: Hero 60 HP, Goblin 38 HP
Round 6: Hero 52 HP, Goblin 33 HP
Round 7: Hero 44 HP, Goblin 30 HP
Round 8: Hero 36 HP, Goblin 26 HP
Round 9: Hero 28 HP, Goblin 20 HP
Round 10: Hero 20 HP, Goblin 18 HP
Round 11: Hero 12 HP, Goblin 16 HP
Round 12: Hero 4 HP, Goblin 9 HP
Round 13: Hero 0 HP, Goblin 7 HP
Goblin wins after 13 rounds!
```

Discussion

- **Encapsulation:** `_hp` is only changed by the character itself, through `take_damage()` and `heal()`. `attack()` calls the target's `take_damage()`, and `hp` and `is_alive` are read-only properties.
- **Composition:** A `Weapon` has a `Dice`, and a `Character` has a `Weapon`. A character without a weapon uses its default `damage`.
- **Inheritance:** `Warrior` overrides `take_damage()` to subtract armor. `Mage` overrides `attack()` to spend 10 mana on a double-damage spell, and otherwise attacks normally and restores 5 mana.
- **Battle:** `play_round()` lets the hero attack first, and the enemy only attacks back if it is still alive. `simulate_battle()` repeats this until one character dies.
- **Randomness:** The results vary with the dice rolls. We use `random.seed(42)` so the notebook gives the same output each time.

Use of GenAI

Tool: Claude (Anthropic).

What we used it for

- **Project setup:** how to organise the repository, how to upload our Task 1-3 files to GitHub from our own accounts, how to clone the repository, and how to create the notebook in VS Code with a new virtual environment.
- **Compatibility check:** we shared our `Character`, `Dice` and `Weapon` classes so that the code for Task 4 and 5 would build on our own `Character` class (`take_damage()`, `is_alive`, `weapon`) without changing it.
- **Task 4 and 5:** `Warrior`, `Mage` and `Battle` are based on code suggested by Claude, written to fit our class design.
- **Notebook:** Claude helped with the structure, the helper function that presents the classes, and the discussion text.

What we did ourselves

- We wrote Tasks 1-3 together (`Character`, `Dice`, `Weapon`), including our own tests, and Claude did not change this code.
- We ran the notebook from top to bottom and checked that the battles worked and that the results made sense.